



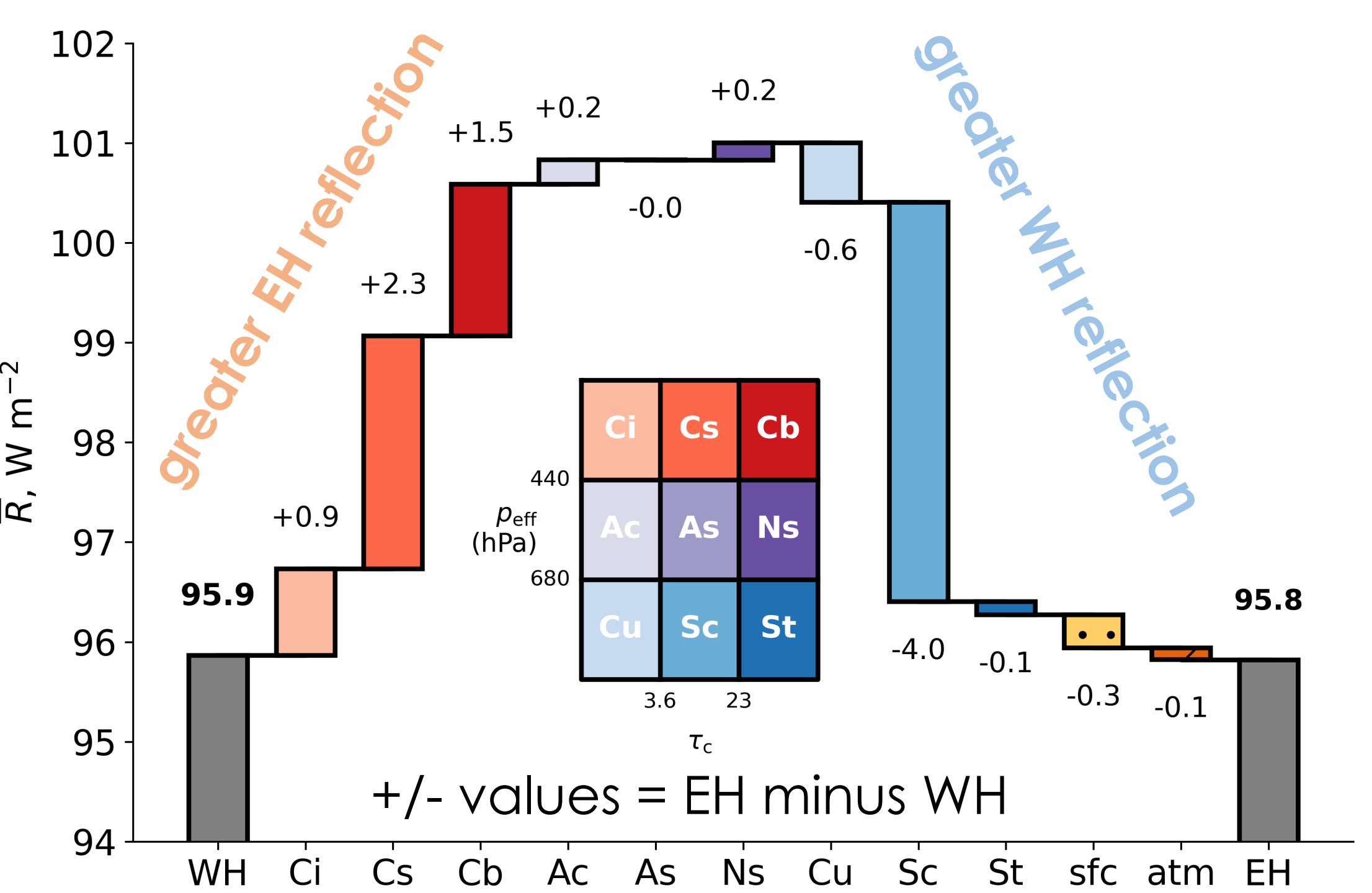
Earth's East-West Albedo Symmetry

Jianhao Zhang^{1,2}, Jake Gristey^{1,2,3}, Graham Feingold²

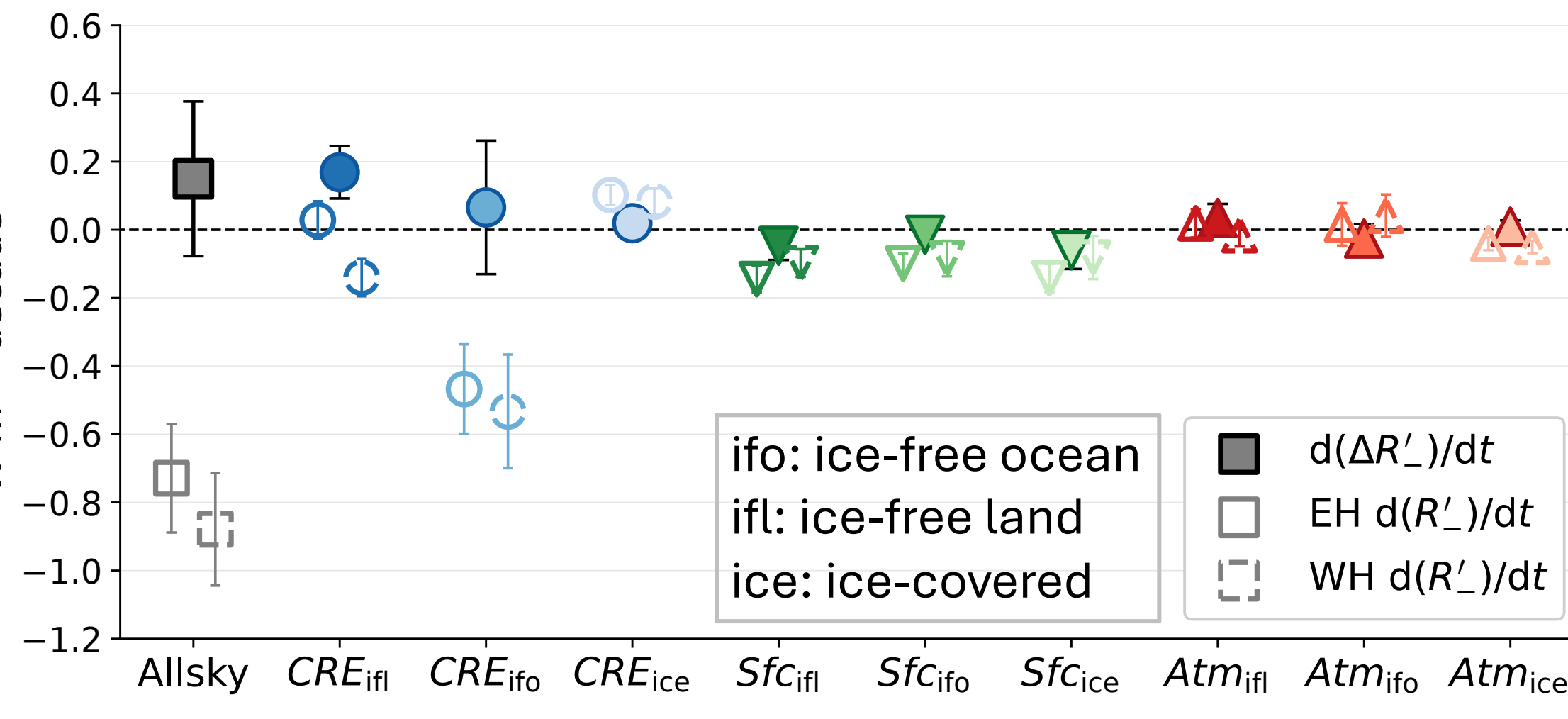
The Discovery:

A unique and persistent East-West albedo symmetry at 27°E, with a “triple symmetry” feature.

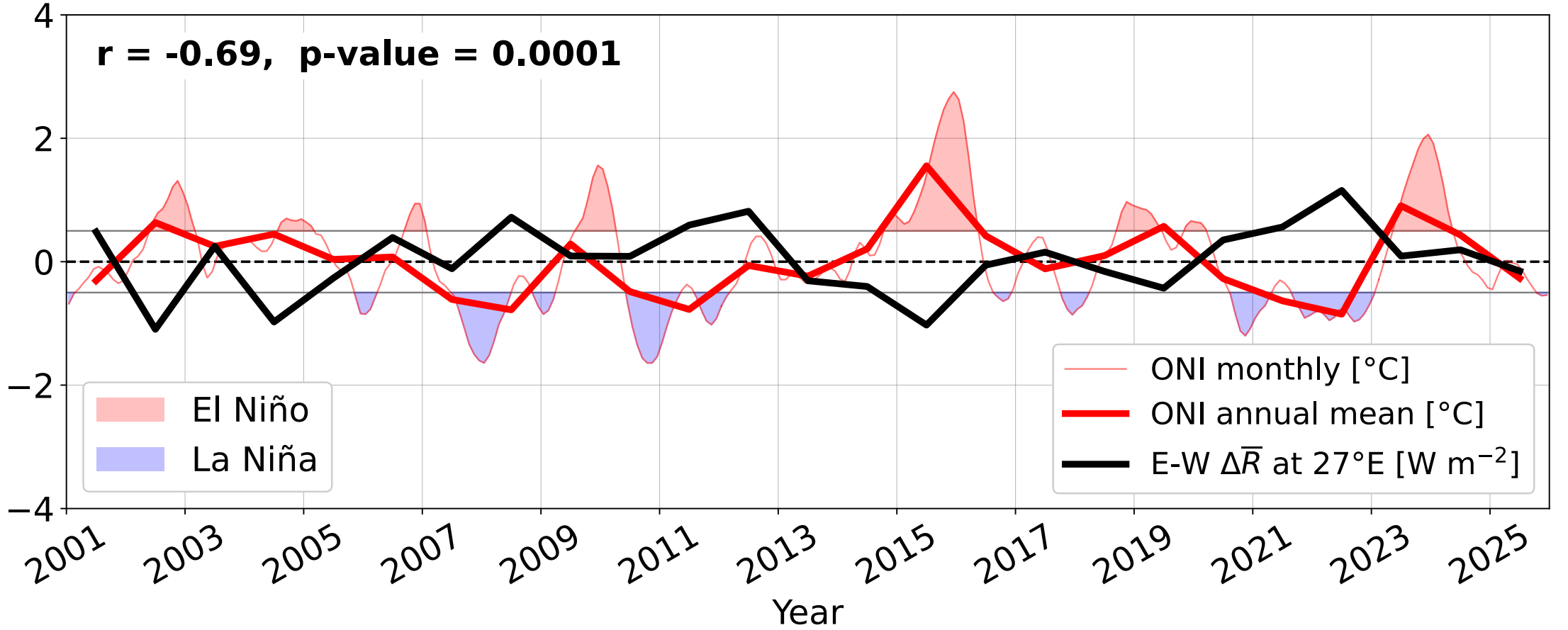
Clouds compensate to sustain symmetry



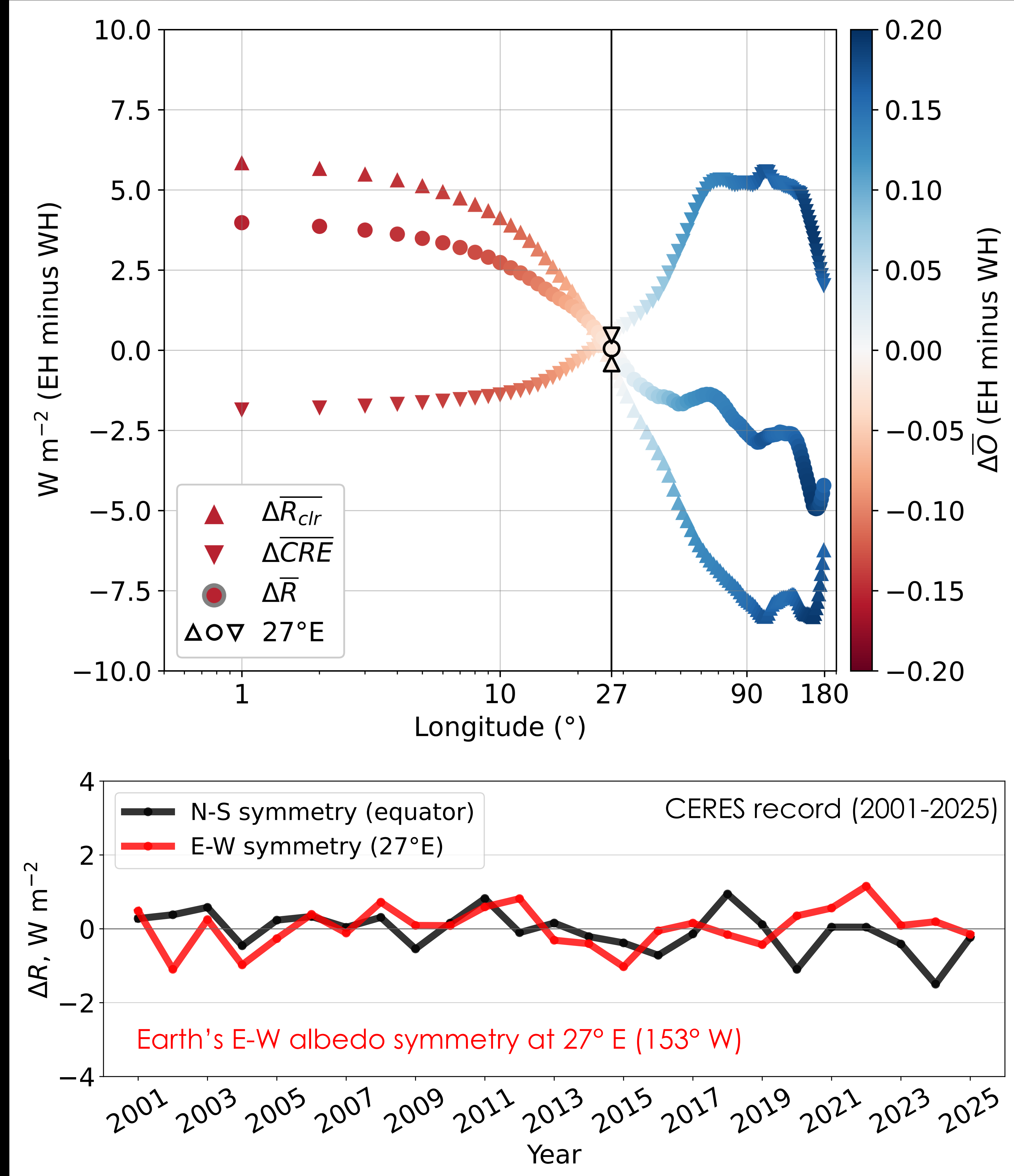
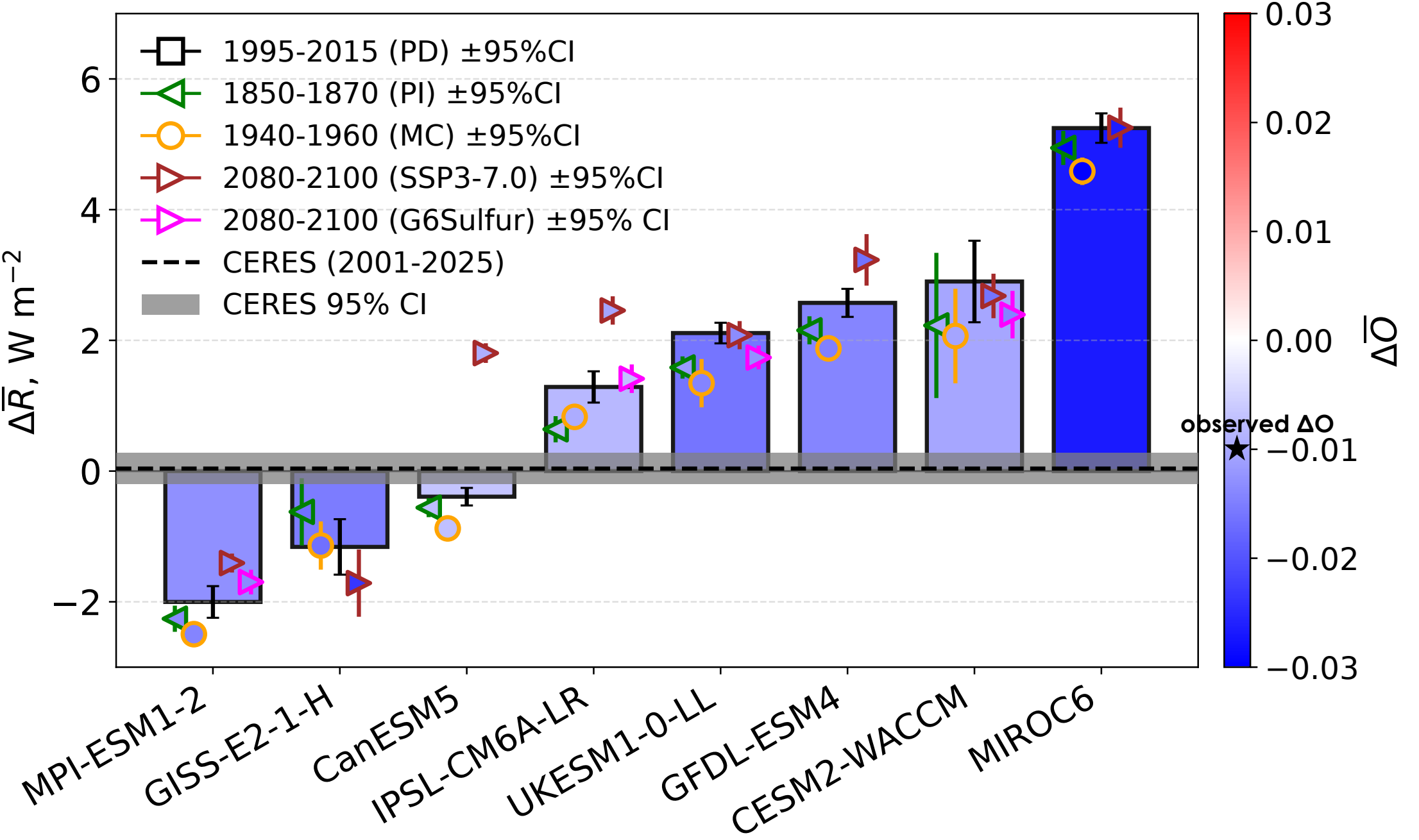
Recent trends towards asymmetry



E-W symmetry tracks ENSO phase



Models do not capture the 27°E symmetry



Zhang, J., Gristey, J. J., and Feingold, G. (2026) Earth's East-West Albedo Symmetry. *Nature*, in press, 10.1038/s41586-026-10624-2.

Why should we care?

Model improvements. The E-W symmetry is a powerful way to evaluate ESMs, with the “triple symmetry” serving as a reduced degree-of-freedom constraint on system couplings.

Fundamental understanding of the Earth system. Cloud feedback, ENSO, and EEI

Implication for SRM. The impacts of Solar Radiation Modification (SRM) cascade through Earth system couplings.

New Dimension, More Questions

Are albedo symmetry pairs fundamentally connected, or do they emerge independently? (E-W and N-S symmetries do not appear connected)

Is the open ocean symmetry essential for the E-W all-sky symmetry?

CERES EBAF & FBCT products
25 years (2001-2025)
R: all-sky SW (0.3-5 μm) reflection at TOA
R_{clr}: TOA SW reflection assuming clear-sky
CRE: Cloud Radiative effect
CRE = R - R_{clr}